



Research paper

Ship Collision Risk Evaluation using AIS and weather data through fuzzy logic and deep learning

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ABSTRACT

According to the European Maritime Safety Agency, marine casualties and incidents are mainly caused by 'Human action' which highlights the need for more advanced technologies to enhance the safety of ships, particularly through improved risk assessment of ship collisions. Accordingly, this study explores the application of Deep Learning for Ship Collision Risk Evaluation to detect ship collisions. A Collision Risk Index (CRI) between two vessels at a given instance is calculated using AIS data. The environmental impact is captured using an Environmental Factor (Enfactor) index for own and target ships. These indices are combined using different weights to obtain a Resultant that represents collective impact of weather factors on both the ships. Subsequently, CRI and Resultant Enfactor indices are combined using Fuzzy Logic to obtain a comprehensive index called Enhanced Collision Risk Index with Weather (ECRI-W). Various deep learning models are evaluated and the Self-Attention and Intersample Attention Transformer (SAINT) model is selected due to its superior performance with tabular data. The Piraeus AIS and Weather datasets are used for model training and testing. After preprocessing the datasets, the model undergoes self-supervised pretraining and supervised finetuning. It is concluded that SAINT-s variant with pretraining achieves the best performance.

1. Introduction

The shipping industry today accounts for transporting approximately 90% of the world's goods, including raw materials, finished products, and essential commodities like food and energy supplies (International Chamber of Shipping, 2024). Ships connect distant economies, allowing countries to participate in global trade and access goods and services they cannot produce locally. Like all other transportation industries, the shipping industry is also grappling with 3 main challenges: cost, safety, and sustainability. These challenges can be looked at as key objectives in the engineering design and operations of ships. Significant efforts and investments have been put into making shipping safe, cost efficient and sustainable for the environment.

However, according to the European Maritime Safety Agency (EMSA), 23,814 marine casualties and incidents related to European interests took place between 2014 and 2022 with an annual average of 2646 (EMSA, 2024). Of these, 45.3% took place while in port areas, i.e., in departure, arrival and anchored or alongside, and 44% took place during the enroute (mid-water or transit) segment. The predominant cause of these events was due to 'Human action', accounting for 59.1% of the total number of events. These statistics prove that despite having sophisticated technologies like Radio Detection and Ranging

(RADAR) and Electric Chart Display and Information System (ECDIS), accidents still occur and majority of them do so due to human errors. Hence, more effective technologies are needed to not only help with navigation but also with the risk assessment of ship collisions. In recent years, there has been a significant upward trend in the interest in the application of Artificial Intelligence (AI) to the shipping industry. This interest is expected to grow as new algorithms are developed and new avenues for application are identified (Xiao et al., 2024).

Application of AI in the shipping industry mainly fall under four categories. Predictive Analytics and Forecasting is used to address topics such as Ship Trajectory Prediction (Jaskólski, 2017; Fossen and Fossen, 2018; Sekhon and Fleming, 2020; Wang et al., 2023), Ship Fuel Prediction (Zhang et al., 2024; Bocchetti et al., 2013; Hu et al., 2019; Yan et al., 2020; Chen et al., 2023), Prediction of Hull girder loads during ship motion (Wang et al., 2024b), Predictive maintenance (Park and Oh, 2023), Prediction of corrosion (Imran et al., 2023) etc. Computer Vision has been applied for Drowning Person Detection (Shi et al., 2024), Ship Engine Room Equipment Detection (Chen et al., 2024), to carry out review of ship safety plans using Object Detection method (Kong et al., 2022) etc. Robotics has been applied to address biofouling wherein an autonomous system which can recognize and

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